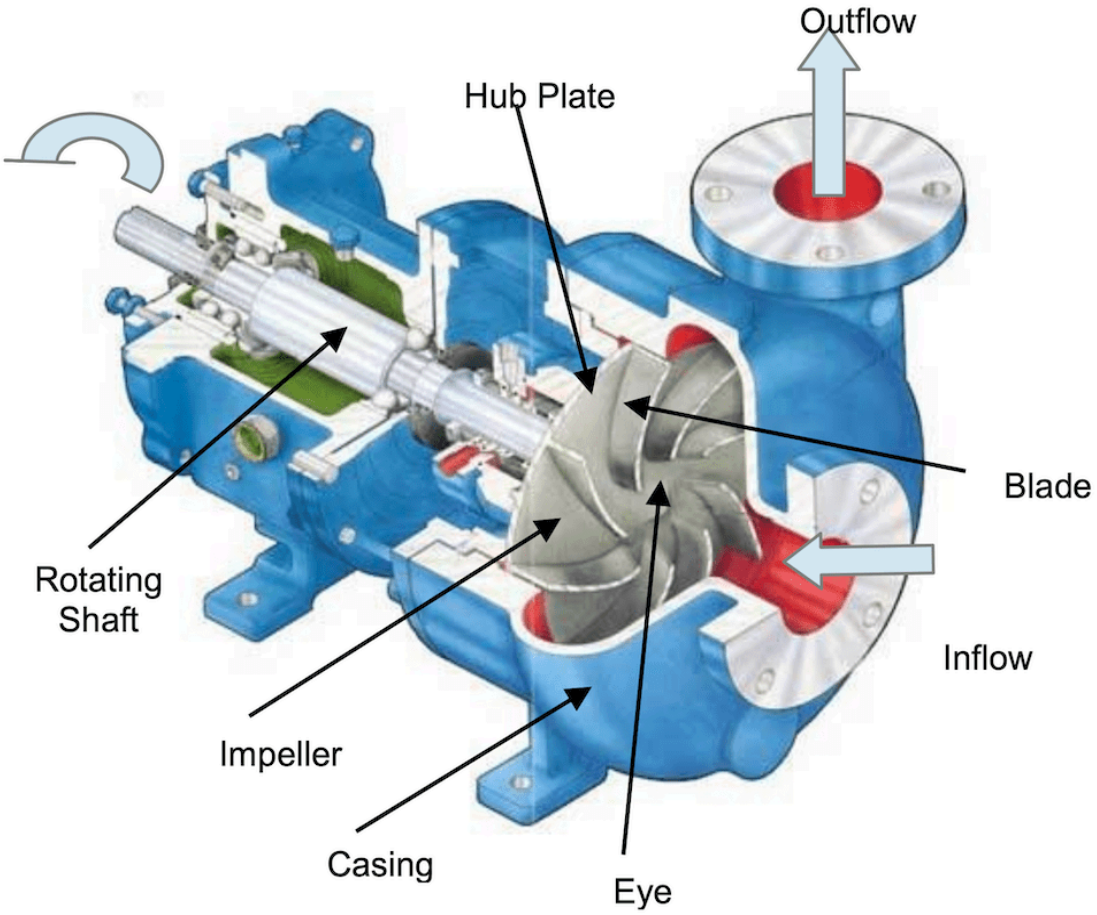
CENTRIFUGAL PUMP AND NPSH ANALYSIS:

A SIMULATION REPORT

Operating Point and Cavitation Assessment Using Excel and DWSIM.



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1 Introduction

A pump is a device used to transport fluid from one place to another. It works by pushing the fluid by the use of an impeller. One common example is the borehole pump in many homes which raises water from an underground reservoir to a storage tank. On a much larger scale, process plants such as oil and gas facilities use pipeline networks to move products between vessels. Pumps supply the energy that keeps the liquid flowing. Improperly selected or operated pumps can waste energy, deliver the wrong flow, or fail early, so designing and operating them correctly matters.

1.1 The Problem

- *What controls the flow a pump delivers on a given pipe system?*
- *How does a pump cavitate? When does a pump cavitate?*

To answer the following questions, we need to consider not just the pump but also the system in which it is being used. This report answers both questions for a typical water-transfer system, based on a manufacturer's pump curve.

1.2 Objectives and approach:

- ❖ To find the operating point of the pump on the pipe system.

- To show how a dirty pipe and a higher static head shift the operating point.
- To verify the Excel results in DWSIM.
- To assess cavitation risk with NPSHa vs NPSHr, including temperature.
- To identify and rank the failure modes (FMEA).

The approach was to build the system curve and a fitted pump curve in Excel, find where they meet, and check the result in a DWSIM simulation. Section 2 gives the theory, Section 3 describes the system and method, Section 4 presents the operating-point results, Section 5 checks them in DWSIM, Section 6 assesses cavitation, Section 7 gives the FMEA and Section 8 concludes.

2 Theory

Two curves decide where a pump will operate in a pipe system: the system curve and the pump curve. The operating point is where they meet.

2.1 System curve

The system curve shows the head the pipe system needs at each flow rate. This head has two parts: the static head, which is the vertical lift between the liquid levels and does not change with flow, and the friction head, which grows with flow.

$$H_{sys} = H_s + H_f \quad (1)$$

The friction head comes from the Darcy-Weisbach equation[1]:

$$H_f = f \cdot (L/D) \cdot v^2/(2g) \quad (2)$$

Velocity is related to flow by:

$$v = Q/A \quad (3)$$

Substituting (3) into (2):

$$H_f = [\, f \cdot L \,/\, (2g \cdot D \cdot A^2) \,] \cdot Q^2 = kQ^2 \quad (4)$$

where H_{sys} is the system head (m), H_s is the static head (m), H_f is the friction head (m), f is the friction factor, L is the pipe length (m), D is the pipe diameter (m), A is the pipe cross-sectional area (m²), g is gravitational acceleration (m/s²), Q is the flow rate and k is the system friction constant. In the equations above Q is in m³/s. In Excel it is in m³/h, so k includes a unit conversion.

k collects the friction factor and the pipe geometry: $k = fL/(2gDA^2)$. It is treated as constant because f changes very little in turbulent flow. k gets bigger with a longer pipe, a smaller diameter, rough or dirty pipe walls, and more valves and fittings. Because friction goes with Q², doubling the flow makes the friction head four times larger.

2.2 Pump curve

The pump curve shows the head the pump can supply at each flow rate. The head falls as the flow rises, so the curve slopes downward. It is supplied by the manufacturer in the datasheet [2], along with the efficiency and NPSHr curves.

Points were read from the manufacturer's curve [2] and plotted in Excel. A third-order polynomial trendline was then fitted to them:

$$H_{pump} = a_0 + a_1 Q + a_2 Q^2 + a_3 Q^3 \quad (5)$$

where H_{pump} is the pump head (m), Q is the flow rate (m³/h) and a_0 to a_3 are the fitted coefficients. a_0 is the shutoff head, the head at zero flow. The pump curve is fairly flat near zero flow and then drops faster. A lower-order fit cannot follow this bend, and a higher-order fit starts chasing small reading errors, so a third-order fit was used. The fitted equation is $H_{pump} = 15.6 - 0.00835Q + 0.00039Q^2 - 0.00000996Q^3$, with Q in m³/h, and the fit has $R^2 = 1.00$. It is only valid within the flow range of the datasheet, so it is not extrapolated beyond it.

2.3 Operating point

The operating point is the flow (Q_{op}) and head $(H_{op})$where the pump curve and the system curve meet:

$$H_{pump}(Q) = H_{sys}(Q) \quad (6)$$

This was solved in Excel with Goal Seek. At flows below Q_op the pump supplies more head than the pipe needs, so the water speeds up. Above Q_{op} The pipe needs more head than the pump can supply, so the water slows down. The flow settles at Q_{op}, and the head at that point is the head the pump delivers.

If the system curve moves up, because of a higher static head or a bigger k from a dirty pipe, the meeting point slides up the pump curve to a lower flow and a higher head.

2.4 Pressure and head

DWSIM gives pressures, but the pump and system curves are in metres of head. Pressure is converted to head so the two can be compared:

$$H = \Delta P / (\rho \cdot g) \quad (7)$$

where H is the head (m), ΔP is the pressure difference (Pa), ρ is the fluid density (kg/m³) and g is gravitational acceleration (m/s²). For the pump, ΔP is the outlet pressure minus the inlet pressure. Head does not depend on the fluid, but the pressure needed to produce it does. For water, a pressure rise of 1 bar is about 10.2 m of head.

2.5 NPSH and cavitation

Cavitation is the process whereby vapour bubbles formed at a low pressure region collapse back into the impeller. Water doesn't just boil when it's hot! Water boils as soon as its pressure drops to its vapour pressure, which is why the water boils at the top of a mountain. At the pump inlet the pressure is at its lowest, so if it drops to its vapour pressure, vapour bubbles are created which are then pushed up into a higher pressure region by the impeller, where they suddenly collapse. This is the cavitation, which manifests itself by noise, vibration, impeller pitting damage and reduced flow and head [3].
When trying to understand whether a pump will cavitate or not, two quantities are considered. NPSHa (Net Positive Suction Head available) is how far the pressure at the pump inlet sits above the point where the liquid would boil, expressed as head. It depends on the suction side of the system. NPSHr (Net Positive Suction Head required) shows the head margin the pump needs in order to avoid cavitation. It depends on the pump, is listed in the datasheet, and increases when flow increases.

NPSHa is calculated as:

$$NPSHa = (P_s - P_v)/(\rho g) + z_s - H_{f,s} \quad (8)$$

where P_s is the absolute pressure on the surface of the liquid supply (Pa), P_v is the vapour pressure of the liquid at its temperature (Pa), ρ is the density (kg/m³), g is gravitational acceleration (m/s²), z_s is the height of the liquid

surface above the pump inlet (m, negative for a suction lift) and $H_{f,s}$ is the friction head loss in the suction pipe (m).

Cavitation is avoided if:

$$NPSHa > NPSHr \quad (9)$$

In practice NPSHa should exceed NPSHr with a margin, because the NPSHr value is measured under test conditions and the real system is never exactly the same [4]. The margin in this report is checked using both the difference $(NPSHa - NPSHr)$ and the ratio $(NPSHa / NPSHr)$.
Temperature acts mainly through the vapour pressure. When the liquid is heated, P_v increases rapidly, so the first term in equation 8 gets smaller and NPSHa decreases. The pipe, lift and pump don't change, but the water is nearer to boiling, so the cushion is less. That's why it's more difficult to pump a hot liquid than a cold one.

3 System and method

3.1 System description

The system is a water-transfer line made of a supply tank, a suction pipe, a pump and a discharge pipe (Figure 1). The pump is an HRZL IS100-80-125C horizontal centrifugal pump running at 2900 rpm. Its datasheet [2] gives a rated point of 80 m³/h at 12.5 m head, with an efficiency of 64% and an NPSHr of 4.2 m.

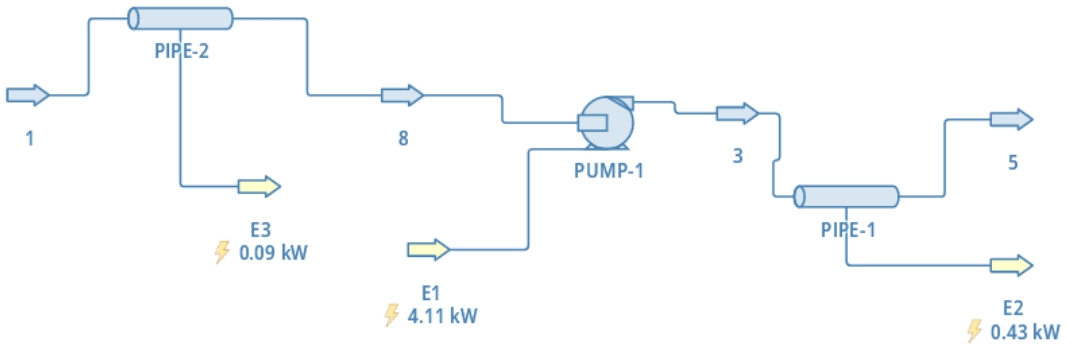


Figure 1: DWSIM flowsheet of the pump system.

Both pipes are raw steel with a 100 mm internal diameter (114 mm external) and a roughness of 4.57×10^{-5} m (Figure 2). The suction pipe is 5 m long and rises 2 m from the tank to the pump. The discharge pipe is 90 m long and rises 6 m from the pump to the delivery point. The total static head is therefore 8 m. The supply and delivery points are at atmospheric pressure, and the base case uses water at 25 °C.

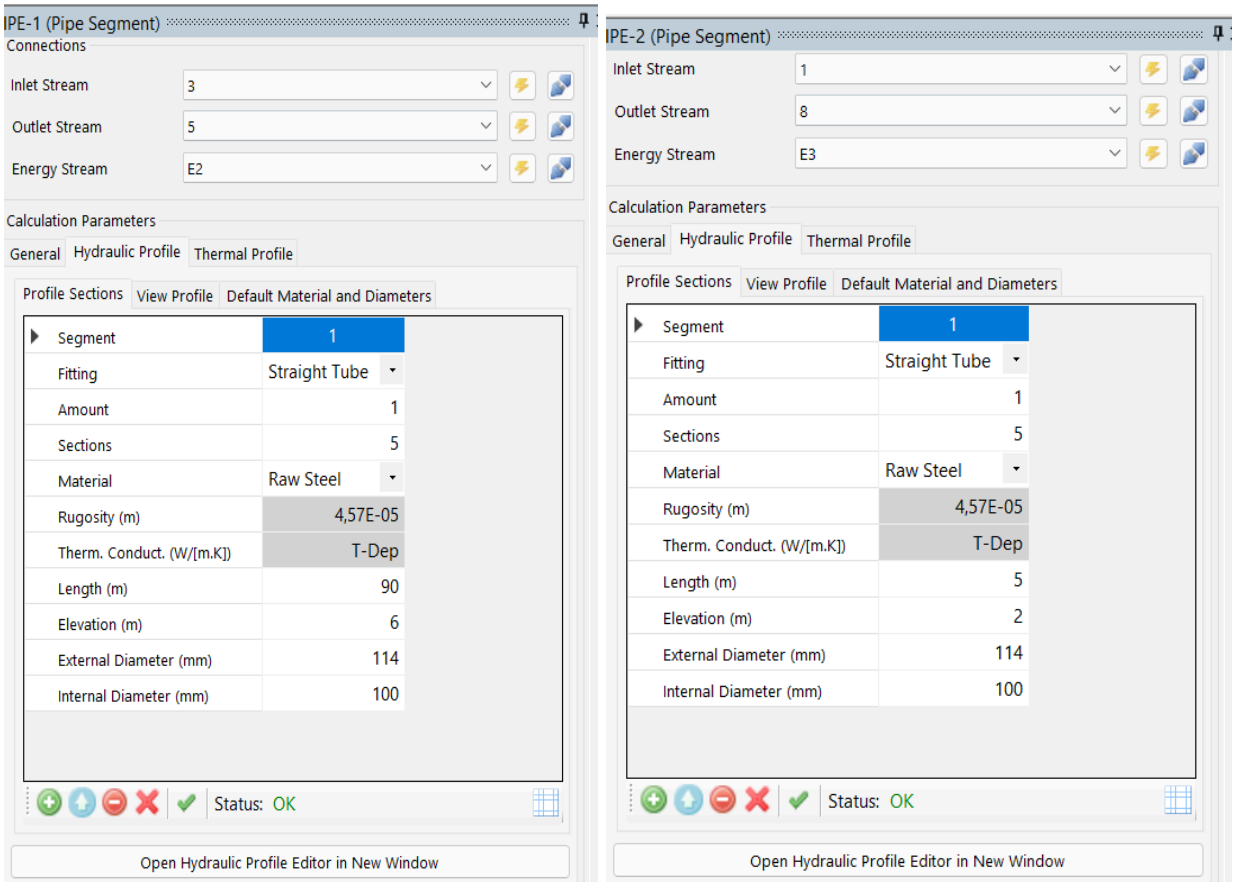


Figure 2: Pipe data in DWSIM. PIPE-1 (left) is the discharge pipe and PIPE-2 (right) is the suction pipe.

3.2 Excel model

The system curve was built from equations 1 to 4 with a static head of 8 m and a friction constant k = 0.001093467 m/(m³/h)²:

$$H_{sys} = 8 + 0.001093467Q^2$$

where Q is in m³/h. The pump curve was the third-order fit of the datasheet points (equation 5). The operating point was found by setting the two equal (equation 6) and solving with Goal Seek. Two more cases were run to see how the operating point moves:

1. Higher static head: H_s raised from 8 m to 9 m, with k unchanged.
2. Dirty pipe: k raised from 0.001093467 to 0.002, with H_s unchanged.

The friction constant k covers the whole line: the 5 m suction pipe and the 90 m discharge pipe, 95 m of 100 mm pipe in all. It was calculated from equation 4 at a reference flow of 71 m³/h, close to the final operating flow. At this flow the velocity is $v = Q/A = 2.51\ m/s\ (A = 0.007854\ m^2)$ and the Reynolds number is $Re = \rho v D/\mu = 2.8\times10^5$, using ρ = 997 kg/m³ and μ = 0.891×10⁻³ kg/(m·s) for water at 25 °C [6], so the flow is turbulent. The relative roughness is $\varepsilon/D = 4.57\times10^{-5} / 0.1 = 4.6\times10^{-4}$. The Colebrook equation [1] then gives f = 0.0181, and equation 4 gives k = 0.001093 m/(m³/h)² (converted from m³/s by dividing by 3600²). Because f depends only weakly on flow, using the final operating flow of 70.6 m³/h instead changes k by less than 0.1%. The suction pipe's share is k_s = k × 5/95 = 0.0000576 m/(m³/h)², used in Section 3.4. In the dirty-pipe case, k = 0.002 corresponds to f ≈ 0.033, a wall roughness of about 0.6 mm (roughly 14 times the clean value), as for a scaled or corroded pipe.

3.3 DWSIM verification

The same system was built in DWSIM [5] (STEAM TABLES IAPWS-IF97 property package) with the two pipes above and the pump's head, efficiency, power and NPSHr curves entered at 2900 rpm (Figure 3). The Excel operating flow of 70.59 m³/h was entered as the feed flow. If the Excel operating point is right, the pump should supply just the head the pipes need, so the water should arrive at the delivery point at atmospheric pressure. Pressures were then read at the pump inlet, the pump outlet and the delivery point, and the pump head was calculated from equation 7. The pump head, the delivery pressure and the NPSH available were then compared with Excel.

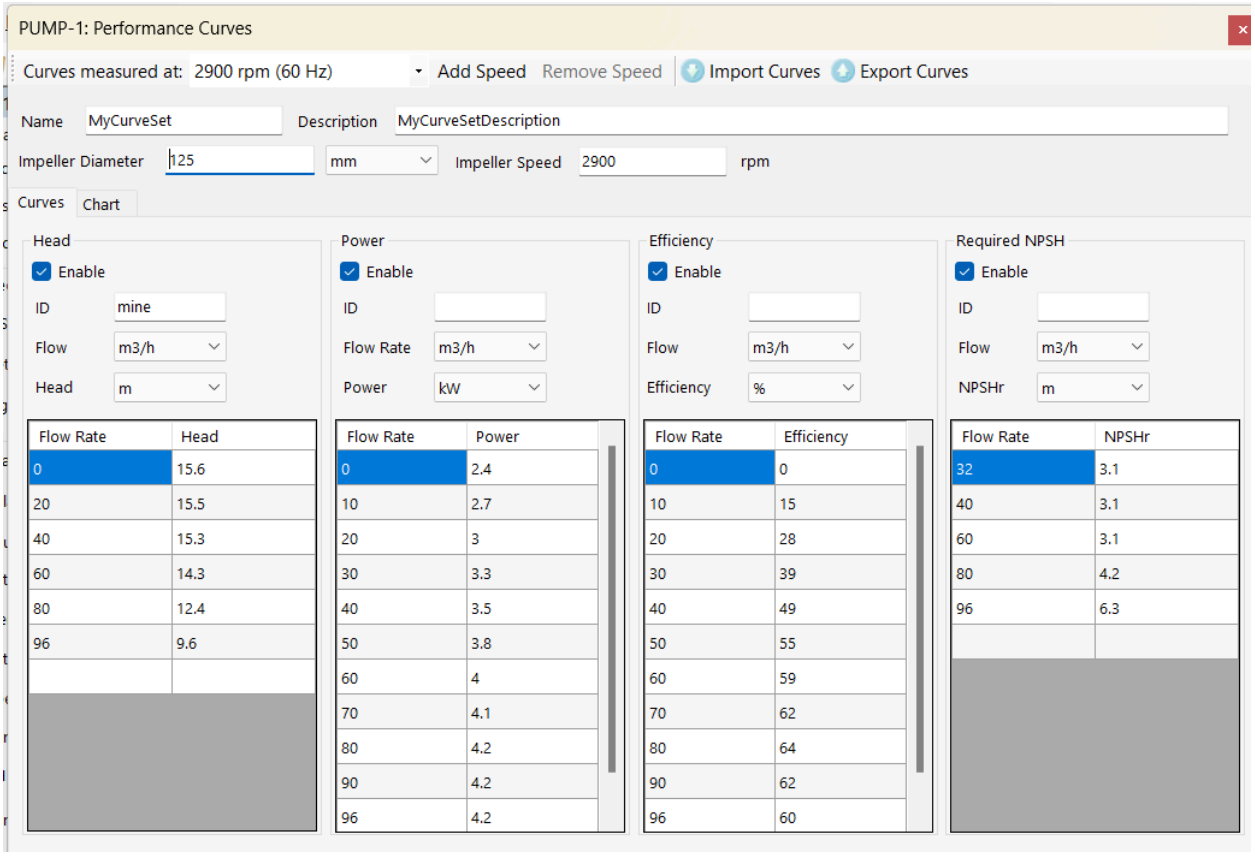


Figure 3: Pump performance curves entered in DWSIM (data from [2]).

3.4 Cavitation assessment

NPSHa was calculated in Excel from equation 8, with P_s = 101.3 kPa, z_s = −2 m (the pump sits 2 m above the supply level) and a suction friction constant of 0.00005755 m/(m³/h)². NPSHr was taken from the datasheet curve, and held constant at 3.1 m below 32 m³/h, where the datasheet curve starts. The check was made at the operating flow for water at 25, 60, 70, 75, 80 and 90 °C, using the vapour pressure and density at each temperature [6]: 3.2 kPa and 997 kg/m³ at 25 °C, 19.9 kPa and 983 kg/m³ at 60 °C, 31.2 kPa and 978 kg/m³ at 70 °C, 38.6 kPa and 975 kg/m³ at 75 °C, 47.4 kPa and 972 kg/m³ at 80 °C, and 70.1 kPa and 965 kg/m³ at 90 °C. NPSHr was linearly interpolated from the datasheet curve at the operating flow. Each temperature was classed by the ratio

NPSHa/NPSHr: Safe if it is at least 1.3, Marginal if it is between 1 and 1.3, and Cavitates if it is below 1.

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4 Results: operating point

4.1 Base case

The pump curve and the system curve meet at 70.6 m³/h and ~13.45 m (Figure 4). Of this head, 8 m is static lift and 5.45 m is friction.

Flow	Pump curve	pipe line
0	15.6	8
8	15.55306048	8.069981871
16	15.52544384	8.279927486
24	15.48655296	8.629836843
32	15.40579072	9.119709944
40	15.25256	9.749546787
48	14.99626368	10.51934737
56	14.60630464	11.4291117
64	14.05208576	12.47883977
72	13.30300992	13.66853159
80	12.32848	14.99818715
88	11.09789888	16.46780645
96	9.58066944	18.07738949

flow	70.59584499
pump curv	13.44993275
pipe line	13.44959039
difference	0.00034236

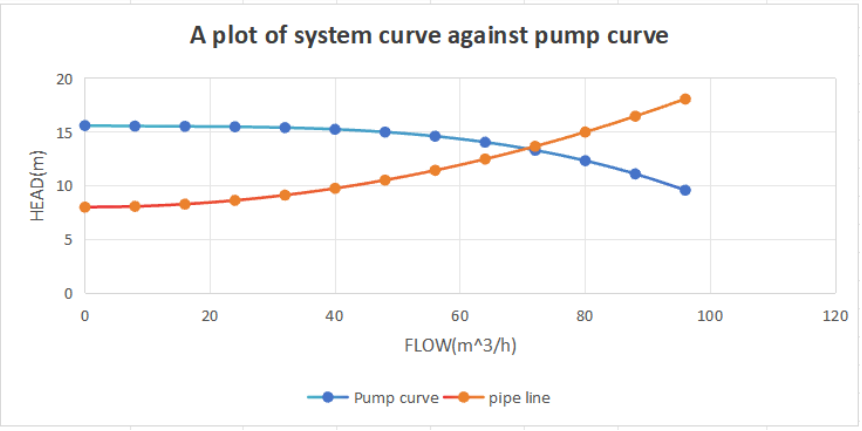


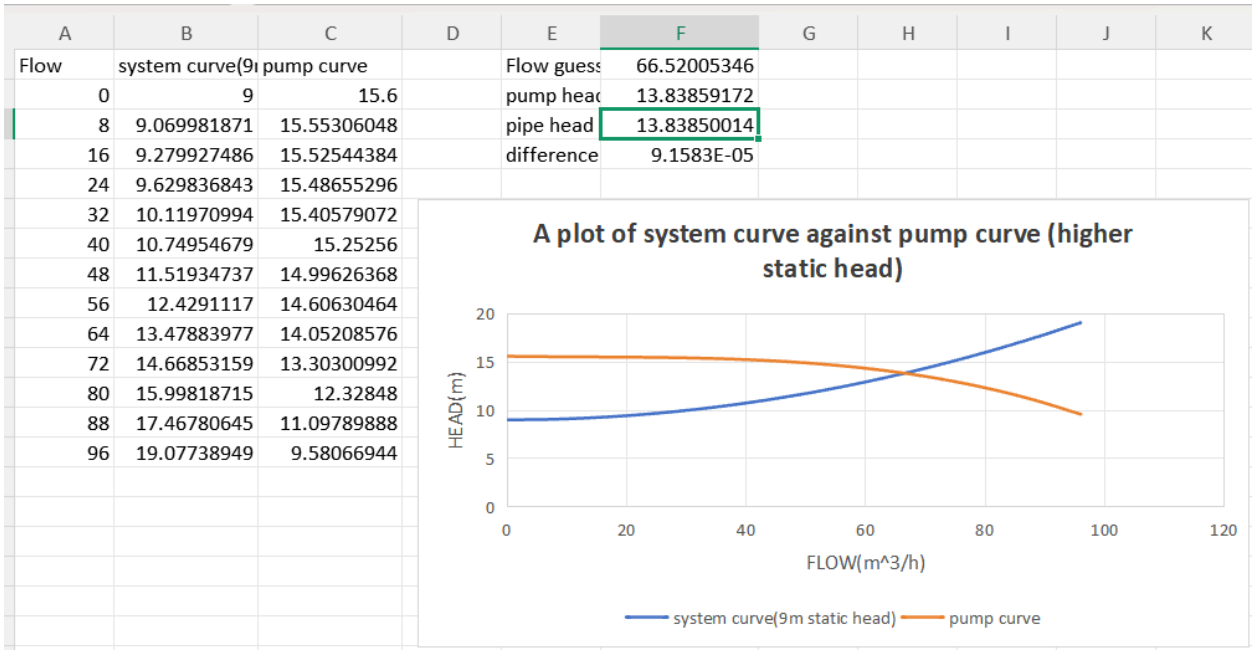
Figure 4: Pump curve and system curve for the base case.

The pump therefore runs to the left of its rated point (80 m³/h, 12.5 m). At 80 m³/h the pipe needs 15.00 m, but the pump can only give 12.33 m, so the flow settles lower, where the two match.

At 70.6m³/h the datasheet efficiency is about 62%, close to the pump's peak of about 64% near 80 m³/h, so the pump is running near its best efficiency point.

4.2 Higher static head

With the static head raised from 8 m to 9 m and k unchanged, the system curve moves up and the operating point slides up the pump curve (Figure 5). The flow drops from 70.6 to 66.52 m³/h, a fall of about 5.8%, and the head rises from 13.45 to 13.84 m.



Flow	system curve(9r	pump curve
0	9	15.6
8	9.069981871	15.55306048
16	9.279927486	15.52544384
24	9.629836843	15.48655296
32	10.11970994	15.40579072
40	10.74954679	15.25256
48	11.51934737	14.99626368
56	12.4291117	14.60630464
64	13.47883977	14.05208576
72	14.66853159	13.30300992
80	15.99818715	12.32848
88	17.46780645	11.09789888
96	19.07738949	9.58066944

Flow guess	66.52005346
pump head	13.83859172
pipe head	13.83850014
difference	9.1583E-05

Figure 5: Operating point with a static head of 9 m.

4.3 Dirty pipe

With k raised from 0.001093467 to 0.002, the friction in the pipe nearly doubles and the system curve gets steeper (Figure 6). The flow drops from 70.59 to about 57.2 m³/h, a fall of ~19%, and the head rises to about 14.5 m.

Flow	dirty pipe,	Pump curve
0	8	15.6
8	8.128	15.55306048
16	8.512	15.52544384
24	9.152	15.48655296
32	10.048	15.40579072
40	11.2	15.25256
48	12.608	14.99626368
56	14.272	14.60630464
64	16.192	14.05208576
72	18.368	13.30300992
80	20.8	12.32848
88	23.488	11.09789888
96	26.432	9.58066944

Flow guess	57.20192213
pump head	14.54408906
pipe head	14.54411979
difference	-3.07291E-05

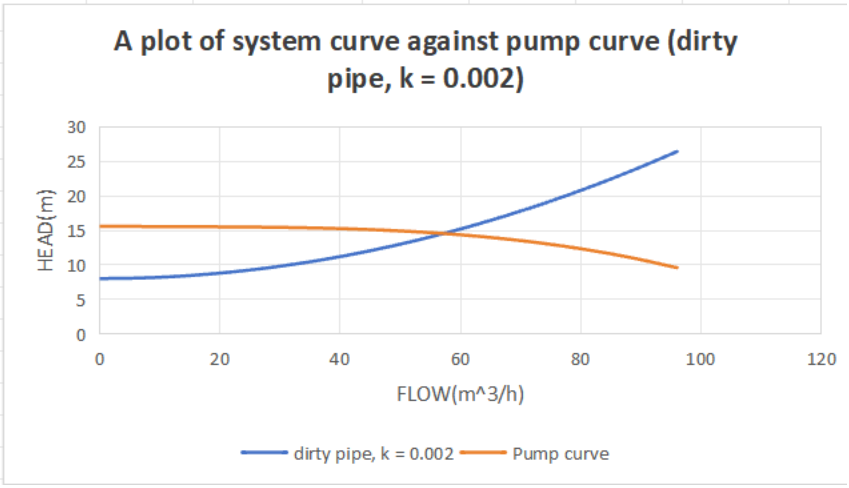


Figure 6: Operating point with a dirty pipe (k = 0.002).

Both cases agree with the theory in 2.3: the system curve moves up, so the operating point moves to a lower flow and a higher head. The dirty pipe has the bigger effect. Friction grows with Q^2, so it makes the curve steeper as the flow rises, while extra static head only lifts the whole curve.

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5 DWSIM verification

The Excel operating flow of 70.59 m^3/h was entered in DWSIM as a feed flow of 19.55 kg/s, which is 0.01961 m^3/s at the pump inlet. With this flow, the pressure at the delivery point (stream 5) comes out at 102,033 Pa, which is around 0.7 kPa above atmospheric pressure (101.3 kPa), or about 0.07 m of head. The pump head DWSIM gives is therefore the head the pipes need, to within 0.07 m (about 0.5% of the pump head), so the DWSIM model confirms the Excel operating

point. The small surplus means DWSIM's own balance point is about 0.3 m^3/h higher than Excel's.

The pump pressure rise is 132,376 Pa, from 78,967 Pa at the inlet to 211,343 Pa at the outlet (Figures 7 and 8). Using the stream density of about 997 kg/m^3 in equation 7, this gives a head of about 13.5 m, within 1% of the Excel value of 13.45 m.

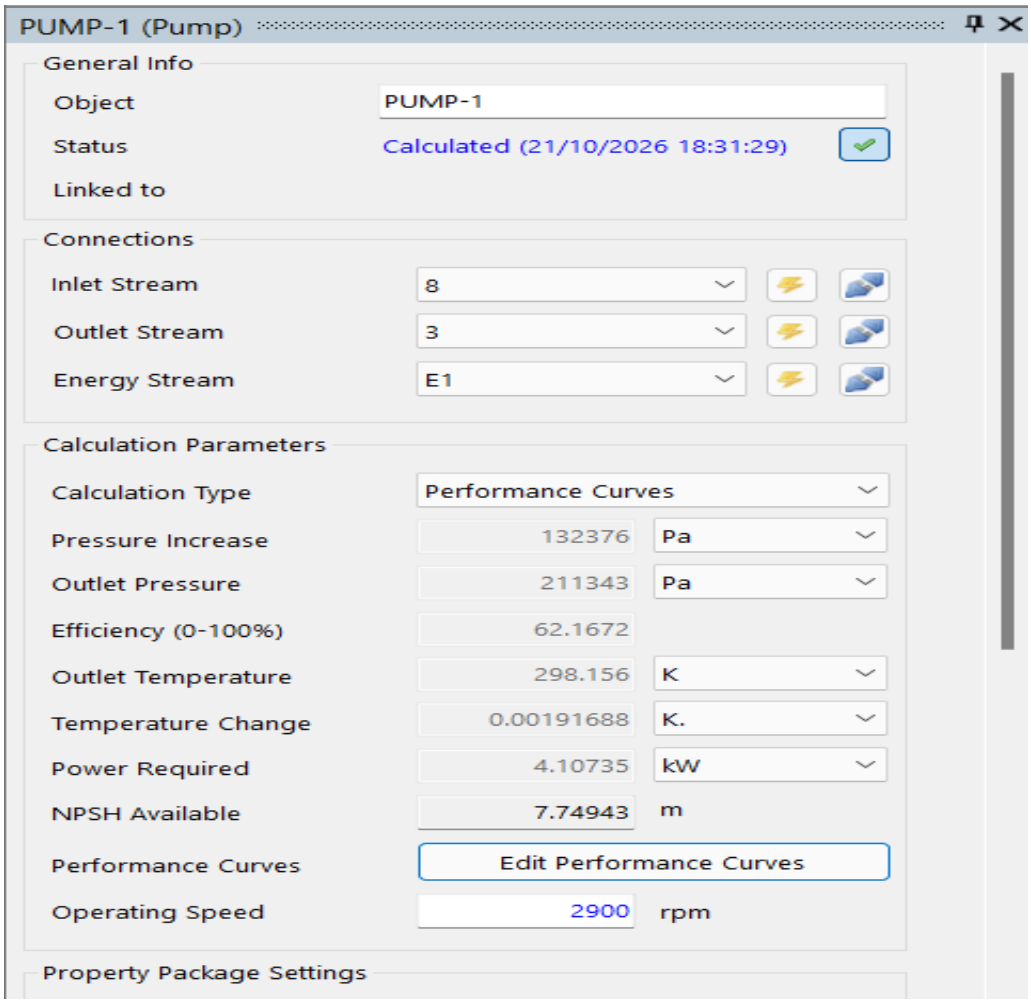
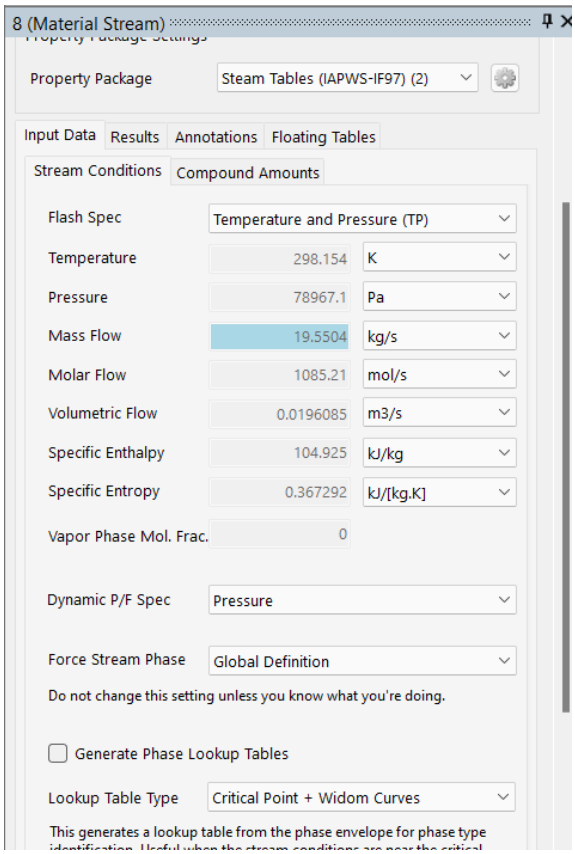
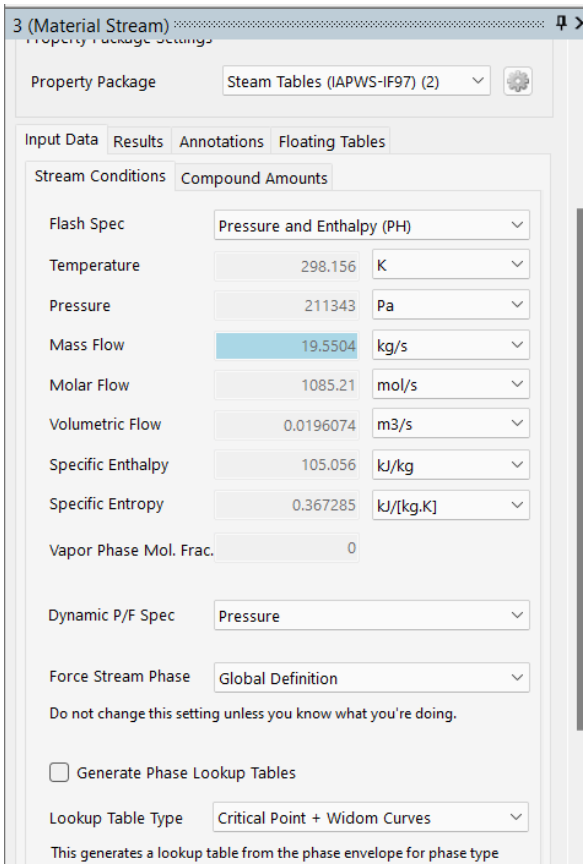


Figure 7: DWSIM pump results.


8 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
Input Data
Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
Temperature and Pressure (TP)
Temperature
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K
Pressure
78967.1
Pa
Mass Flow
19.5504
kg/s
Molar Flow
1085.21
mol/s
Volumetric Flow
0.0196085
m3/s
Specific Enthalpy
104.925
kJ/kg
Specific Entropy
0.367292
kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
Lookup Table Type
Critical Point + Widom Curves
This generates a lookup table from the phase envelope for phase type


3 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
Input Data
Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
Pressure and Enthalpy (PH)
Temperature
298.156
K
Pressure
211343
Pa
Mass Flow
19.5504
kg/s
Molar Flow
1085.21
mol/s
Volumetric Flow
0.0196074
m3/s
Specific Enthalpy
105.056
kJ/kg
Specific Entropy
0.367285
kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
Lookup Table Type
Critical Point + Widom Curves
This generates a lookup table from the phase envelope for phase type

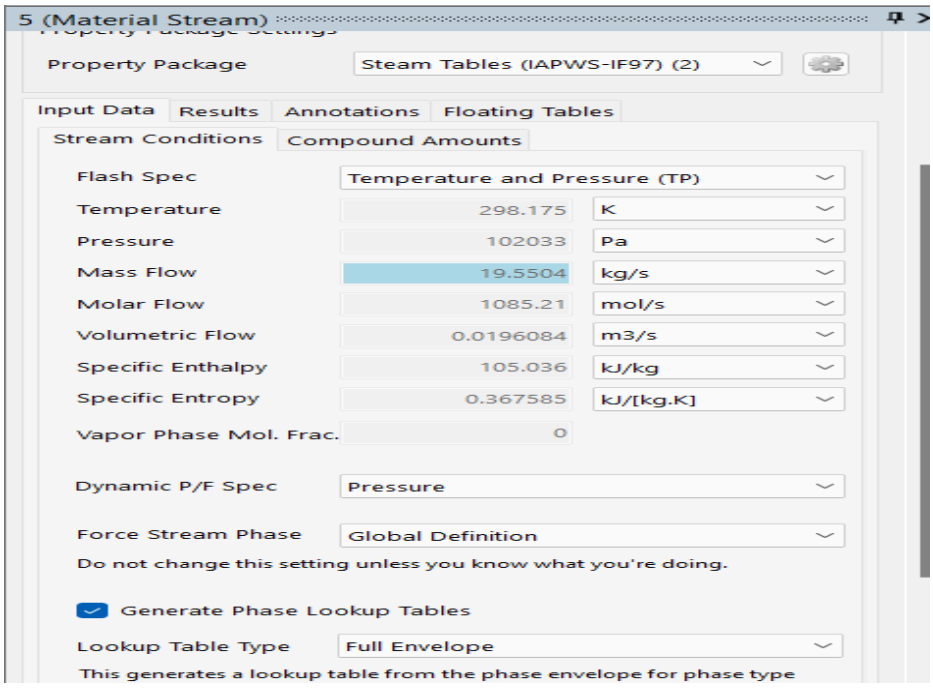

5 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
Input Data
Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
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Temperature
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K
Pressure
102033
Pa
Mass Flow
19.5504
kg/s
Molar Flow
1085.21
mol/s
Volumetric Flow
0.0196084
m3/s
Specific Enthalpy
105.036
kJ/kg
Specific Entropy
0.367585
kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
Lookup Table Type
Full Envelope
This generates a lookup table from the phase envelope for phase type

Figure 8: Pressures at the pump inlet (stream 8), pump outlet (stream 3) and delivery point (stream 5).

The pressures also match the elevations. On the suction side the pressure falls from 101.3 kPa to 79.0 kPa, about 2.3 m of head: the 2 m lift plus about 0.3 m of friction. On the discharge side it falls from 211.3 kPa to 102.03 kPa, about 11.2 m: the 6 m rise plus about 5.2 m of friction in the 90 m pipe. Together this is about 5.5 m of friction in DWSIM against 5.45 m in Excel.

The NPSH available in DWSIM is 7.749 m, against 7.745 m in Excel, a difference of 0.004 m, about 0.05%.

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6 NPSH and cavitation

At the operating flow of 70.6 m³/h, the NPSHr from the datasheet is 3.71 m. NPSHa was calculated for water at six temperatures (Figure 9).

TEMP	pv	rho	NPSHA	margin	ratio	verdict
25	3.2	997	7.745873305	4.035873305	2.08783647	Safe
60	19.9	983	6.156974529	2.446974529	1.659561868	Safe
70	31.2	978	5.022345599	1.312345599	1.353731967	Safe
75	38.6	975	4.271161027	0.561161027	1.151256342	Marginal
80	47.4	972	3.368516963	-0.341483037	0.907956055	Cavitates
90	70.1	965	1.011648246	-2.698351754	0.272681468	Cavitates

Figure 9: NPSHa, margin and ratio at different water temperatures (P_v in kPa, ρ in kg/m³, , both from [6]; NPSHa and margin in m).

At 25 °C, NPSHa is 7.75 m, a margin of 4.04 m (ratio 2.09), so the pump is safe. At 60 °C, NPSHa is 6.16 m (margin 2.45 m, ratio 1.66), still safe, and at 70 °C it is

5.02 m (ratio 1.35). At 75 °C it is 4.27 m (ratio 1.15), which is marginal: above NPSHr, but below the 1.3 ratio used as the margin. At 80 °C it falls to 3.37 m, which is 0.34 m below NPSHr (ratio 0.91), so the pump cavitates. At 90 °C, NPSHa is only 1.01 m (ratio 0.27), far below what the pump needs.

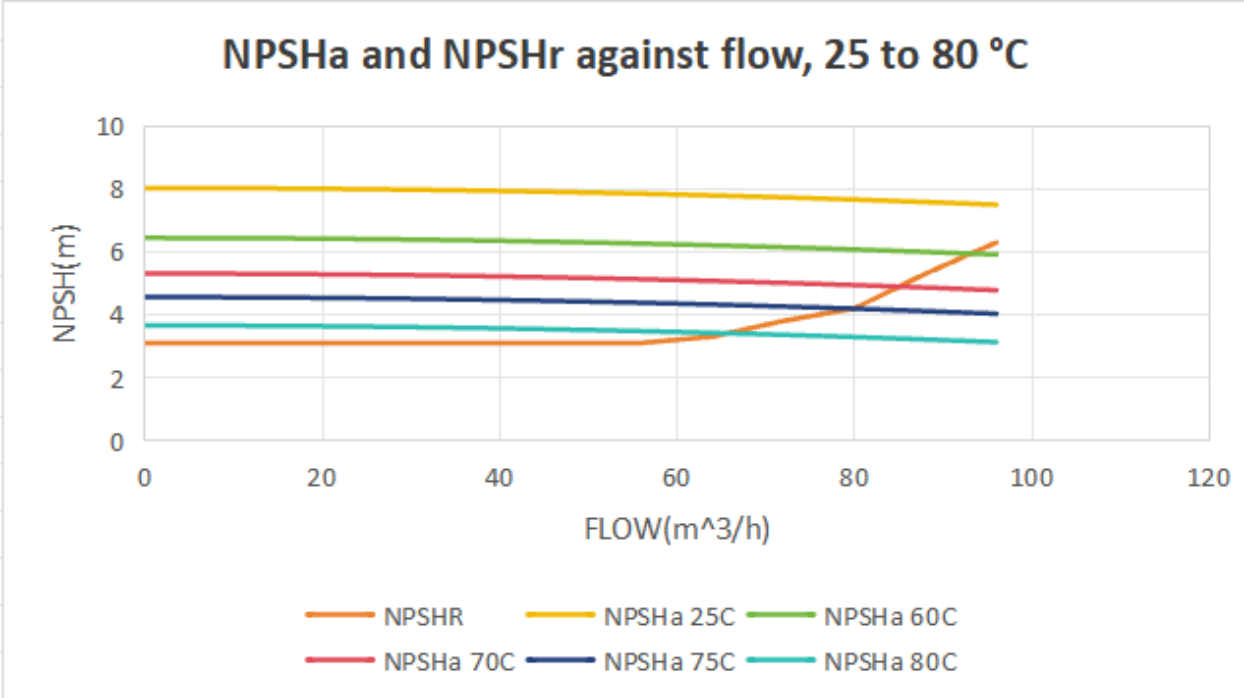


Figure 10: NPSHa at different temperatures against the NPSHr curve.

At 25 °C, NPSHa stays above NPSHr over the whole flow range. At 60 °C and 70 °C, the margin only disappears near the end of the curve, at about 93 and 85 m^3/h. At 75 °C, NPSHa meets NPSHr at about 80 m^3/h, so at the operating flow of 70.6 m^3/h the pump is above NPSHr but with little margin to spare. At 80 °C, NPSHa crosses NPSHr at about 66 m^3/h, so the operating flow of 70.6 m^3/h is already past the crossing and the pump cavitates.

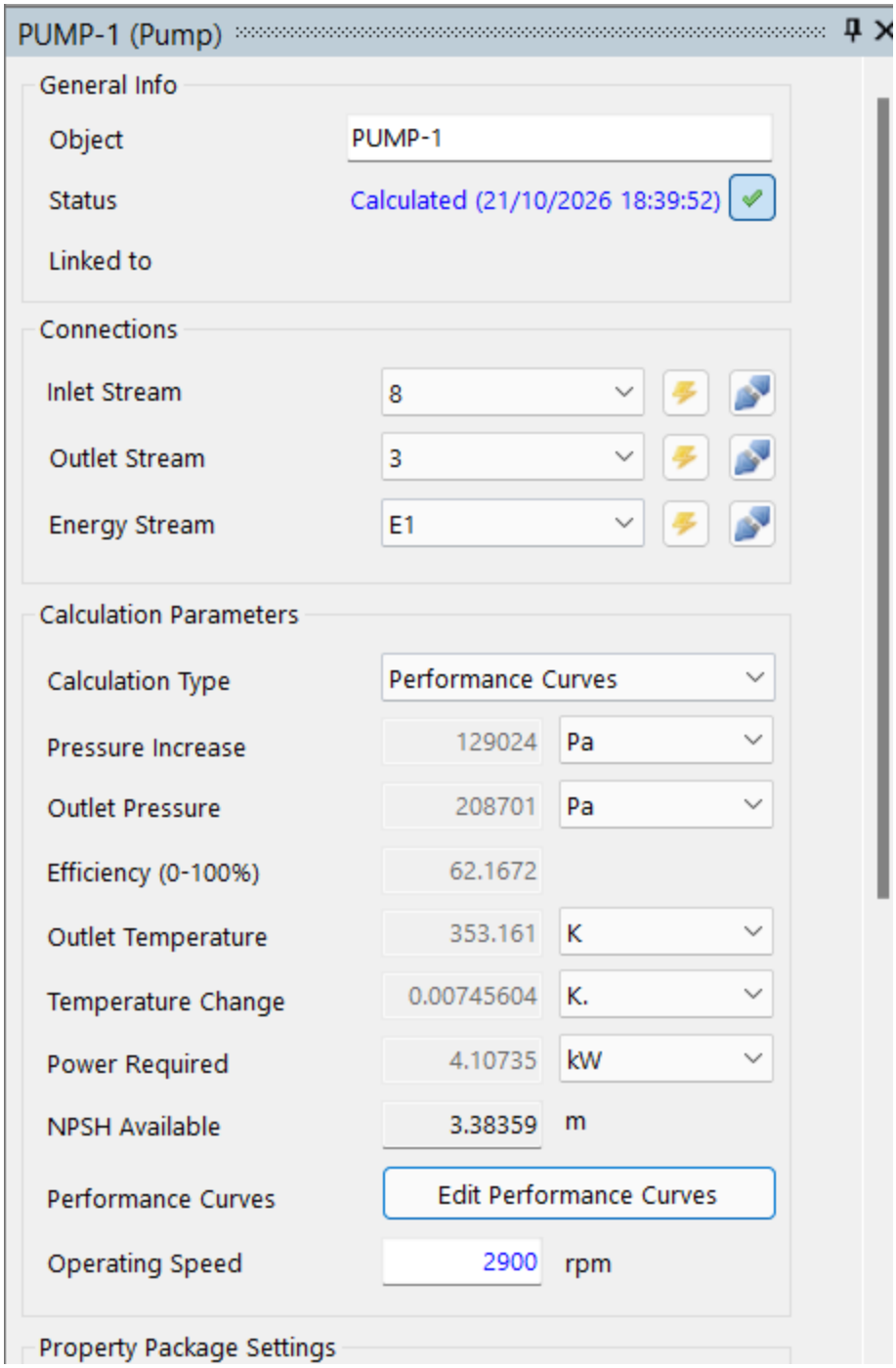


Figure 11: DWSIM NPSH available at 80 °C.

The DWSIM run at 80 °C was kept at the same volumetric flow of 70.59 m³/h, so the mass flow fell from 19.55 to 19.05 kg/s because the water is lighter. At 80 °C, DWSIM gives an NPSH available of 3.38 m. This is close to the Excel value of 3.37

m (a difference of 0.015 m) and is about 0.3 m below the NPSHr of 3.71 m (ratio 0.91). DWSIM and Excel therefore agree that the pump cavitates at this temperature. The pump pressure rise falls from 132,376 Pa to 129,024 Pa because the hot water is lighter (972 kg/m³ against about 997 kg/m³ at 25 °C), but the head is still about 13.5 m, the same as at 25 °C, since the head comes from the pump curve and not from the liquid.

The cause is the vapour pressure. Going from 25 to 80 °C raises it from 3.2 to 47.4 kPa, which removes about 4.4 m from NPSHa. The pipe, the lift and the pump have not changed. The water is just closer to boiling. To avoid cavitation in this situation, cool the water, lower the pump or move it closer to the tank, or run at a lower flow.

7 Failure mode and effects analysis (FMEA)

7.1 Method

An FMEA lists the ways a system can fail, what each failure does and how serious it is [7]. This one covers only the failures the simulation can say something about, so bearing and seal failures are not included. Each failure mode is rated Low, Medium or High for severity (how bad the effect is) and likelihood (how likely the cause is), and Easy, Moderate or Hard for detectability (how easily it is noticed before damage is done). The ratings are engineering judgement, and a qualitative scale was used because no failure-rate data is available for this system. Priority is High when severity is High and likelihood is at least Medium. Within the same priority, modes backed by results from this report rank higher.

7.2 Results

Table 1: Failure modes and the evidence from this report

Failure mode	Cause and effect	Evidence	Control
Cavitation	NPSHa falls below NPSHr (hot water, suction lift, suction friction, high flow). Impeller pitting, noise, vibration, lost head [**3**]	At 25 °C, NPSHa is 7.75 m against NPSHr of 3.71 m (ratio 2.09). At 80 °C it is 3.37 m (ratio 0.91), so the pump cavitates. At the operating flow the limit is about 78 °C, or about 71 °C if a ratio of 1.3 is required	Limit water temperature, keep the suction pipe short, keep a margin, listen for noise and check vibration
Restricted suction pipe or strainer	Debris or scale raises the suction friction, so NPSHa falls and cavitation follows	The suction friction H_f,s lowers NPSHa one for one (equation 8). At 25 °C it would have to rise about 15 times (0.29 m to about 4.3 m) to reach NPSHr, and at 60 °C about 10 times (to about 2.7 m)	Suction pressure gauge, clean the strainer on a schedule
Loss of prime or	Air enters the suction line or the water level drops, so the pump runs	Not modelled. The pump sits 2 m above the supply level, and a standard end-suction centrifugal	Foot valve, airtight suction

dry running	without water and overheats	pump is normally not self-priming (check the datasheet), so it depends on an airtight suction line staying full	pipe, low-level cutoff
Fouled discharge pipe	Scale or corrosion raises friction, so flow falls	In the dirty-pipe case (k = 0.002), flow falls from 70.6 to 57.2 m³/h and head rises from 13.45 to 14.5 m	Compare flow and head against the clean-pipe curve

Table 2: Ratings and priority

Rank	Failure mode	Severity	Likelihood	Detectability	Priority
1	Cavitation	High	Medium (High above about 78 °C)	Moderate: noise and vibration only appear once it has started	High
2	Restricted suction	High (leads to cavitation)	Medium	Moderate: needs a suction gauge	High

3	Loss of prime or dry running	High	Medium (the lift makes it more likely)	Moderate	High
4	Fouled discharge pipe	Medium	Medium	Easy: a drop in flow or head shows against the curve	Medium

7.3 Discussion

Cavitation ranks first because it can damage the impeller and bearings and starts quietly, and because Section 6 shows how close the pump is to it: safe at 60 °C but not at 80 °C. A restricted suction pipe ranks next because it attacks the same margin. Loss of prime is not modelled, but it is a real risk for a pump above the water level. The fouled pipe does no damage and is the easiest to spot, since the flow simply drifts down against the clean-pipe curve.

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8 Conclusion

1. The pump runs at 70.6 m³/h and 13.45 m on the base system, to the left of its rated point (80 m³/h). Of this head, 8 m is static lift and 5.45 m is friction. The efficiency at this flow is about 62%, near the pump's best.
2. A higher static head (9 m) moves the operating point to 66.52 m³/h and 13.84 m, a flow drop of about 5.8%. A dirtier pipe (k = 0.002) moves it to

about 57.2 m^3/h and 14.5 m, a drop of about 19%. Friction has the bigger effect because it grows with Q^2.

3. DWSIM confirms the Excel result. At 70.6 m^3/h the water reaches the delivery point within 0.07 m of head of atmospheric pressure , the pump head is about 13.5 m against 13.45 m, and the NPSH available is 7.749 m against 7.746 m.
4. The pump is safe from cavitation at 25, 60 and 70 °C (NPSHa/NPSHr of 2.09, 1.66 and 1.35), marginal at 75 °C (1.15), and cavitates at 80 °C and 90 °C (0.91 and 0.27). The cause is the vapour pressure, which removes about 4.4 m of NPSHa between 25 and 80 °C. DWSIM agrees at 80 °C (3.38 m against 3.37 m in Excel, both below the NPSHr of 3.71 m).
5. The FMEA ranks cavitation first, followed by a restricted suction pipe and loss of prime (all high priority), with a fouled discharge pipe at medium priority. The first two are backed by the NPSH results in this report.

Limitations: The pump curve points were read by eye from the datasheet, k was treated as constant although the Colebrook friction factor varies from about 0.01905 at 40 m^3/h to 0.0177 at 96 m^3/h, and NPSHr was held at 3.1 m below 32 m^3/h. Excel and DWSIM use the same pump curve, so their agreement checks the pipe side of the model, not the pump data.

References

[1] Y. A. Çengel and J. M. Cimbala, Fluid Mechanics: Fundamentals and Applications, 4th ed. New York, NY, USA: McGraw-Hill, 2018.

[2] Shandong Hairuizhonglian Fluid Technology Co., Ltd. (HRZL), “Performance curve: horizontal centrifugal pump, model IS100-80-125C,” manufacturer datasheet, 2025. [Online]. Available:

https://storage.mtender.gov.md/get/77b6ec40-e2d1-428d-8c9c-30ad6096c138-1756731209099 (accessed Oct. 18, 2026).

[3] I. J. Karassik, J. P. Messina, P. Cooper, and C. C. Heald, Eds., Pump Handbook, 4th ed. New York, NY, USA: McGraw-Hill, 2008.

[4] Rotodynamic Pumps – Guideline for NPSH Margin, ANSI/HI 9.6.1-2017, Hydraulic Institute, Parsippany, NJ, USA, 2017.

[5] D. Medeiros, “DWSIM – Open source chemical process simulator,” DWSIM Patreon Edition, version 8.8.0. [Online]. Available: https://dwsim.org (accessed Oct. 18, 2026).

[6] Y. A. Çengel and J. M. Cimbala, “Table A–3: Properties of saturated water,” in Fluid Mechanics: Fundamentals and Applications, McGraw-Hill, [Online]. Available: https://www.me.psu.edu/cimbala/me405/Links/Table_A_3_CC_Saturated_Water.pdf (accessed Oct. 18, 2026).

[7] Failure Modes and Effects Analysis (FMEA and FMECA), IEC 60812:2018, International Electrotechnical Commission, Geneva, Switzerland, 2018.

[8] Cover image source: https://www.simscale.com/blog/what-is-centrifugal-pump/